



INTERNATIONAL CYBERSECURITY AND DIGITAL FORENSICS ACADEMY

ASSIGNMENT

Assignment Title: Number Systems, ASCII, Timestamps and Data Representation

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Course Title: Basic Computer Skills for Digital Forensics

Course Code: SBT-DF201

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INTRODUCTION

The goal of this laboratory exercise is to build foundational command-line discipline, evidence documentation habits, and repeatable workflow steps using the Windows Command Prompt. Rather than investigating an active security incident, this lab establishes core competencies in navigating the Windows file system, managing evidence directories, executing basic text searches, collecting hardware/network/process triage artifacts, and developing reusable automation batch scripts.

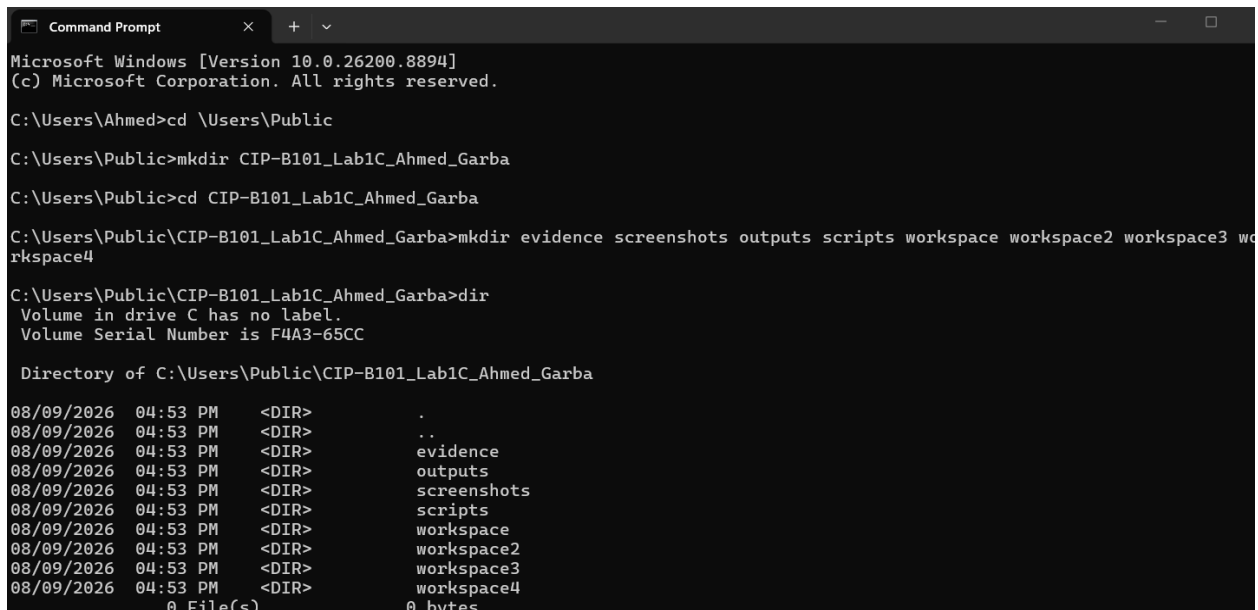
Environment

- **Operating System:** Windows 11 Host / Training Virtual Machine
- **Execution Interface:** Windows Command Prompt (cmd.exe) and Notepad
- **User Privilege Level:** Standard User / Administrator (for PATH configuration)
- **Lab Workspace Directory:** C:\Users\Public\CIP-B101_Lab1C_Ahmed_Garba

Methodology

Establishing the Evidence Workspace

A dedicated, isolated workspace was established within the public user directory to ensure that all artifacts, evidence files, scripts, and logs remained neatly organized and partitioned from critical operating system files.



```
Microsoft Windows [Version 10.0.26200.8894]
(c) Microsoft Corporation. All rights reserved.

C:\Users\Ahmed>cd \Users\Public

C:\Users\Public>mkdir CIP-B101_Lab1C_Ahmed_Garba

C:\Users\Public>cd CIP-B101_Lab1C_Ahmed_Garba

C:\Users\Public\CIP-B101_Lab1C_Ahmed_Garba>mkdir evidence screenshots outputs scripts workspace workspace2 workspace3 workspace4

C:\Users\Public\CIP-B101_Lab1C_Ahmed_Garba>dir
Volume in drive C has no label.
Volume Serial Number is F4A3-65CC

Directory of C:\Users\Public\CIP-B101_Lab1C_Ahmed_Garba

08/09/2026  04:53 PM    <DIR>          .
08/09/2026  04:53 PM    <DIR>          ..
08/09/2026  04:53 PM    <DIR>          evidence
08/09/2026  04:53 PM    <DIR>          outputs
08/09/2026  04:53 PM    <DIR>          screenshots
08/09/2026  04:53 PM    <DIR>          scripts
08/09/2026  04:53 PM    <DIR>          workspace
08/09/2026  04:53 PM    <DIR>          workspace2
08/09/2026  04:53 PM    <DIR>          workspace3
08/09/2026  04:53 PM    <DIR>          workspace4
               0 File(s)              0 bytes
```

Task 1: Windows Paths and Navigation

To understand file-system layout and evidence localization, directory traversal was performed using both absolute and relative paths.

- **Display current location:**

```
C:\Users\Public\CIP-B101_Lab1C_Ahmed_Garba>cd
C:\Users\Public\CIP-B101_Lab1C_Ahmed_Garba

C:\Users\Public\CIP-B101_Lab1C_Ahmed_Garba>echo Current directory is: %cd%
Current directory is: C:\Users\Public\CIP-B101_Lab1C_Ahmed_Garba

C:\Users\Public\CIP-B101_Lab1C_Ahmed_Garba>
```

Navigate to root and specific paths:

```
08/09/2026 04:53 PM <DIR>      evidence
08/09/2026 05:06 PM <DIR>      outputs
08/09/2026 04:53 PM <DIR>      screenshots
08/09/2026 04:53 PM <DIR>      scripts
08/09/2026 04:56 PM <DIR>      workspace
08/09/2026 04:58 PM <DIR>      workspace2
08/09/2026 04:57 PM <DIR>      workspace3
08/09/2026 04:53 PM <DIR>      workspace4
          0 File(s)          0 bytes
          10 Dir(s) 141,826,125,824 bytes free

C:\Users\Public\CIP-B101_Lab1C_Ahmed_Garba>cd \Users
C:\Users>cd Public
C:\Users\Public>cd CIP-B101_Lab1C_Ahmed_Garba
C:\Users\Public\CIP-B101_Lab1C_Ahmed_Garba>tree
Folder PATH listing
Volume serial number is 000000F3 F4A3:65CC
C:.
|_ evidence
|_ outputs
|_ screenshots
|_ scripts
|_ workspace
|_ workspace2
|_ workspace3
|_ workspace_backup
```

Task 2: Basic File and Folder Creation

Evidence note creation and file content generation were practiced inside the designated workspace subdirectory.

```
C:\Users\Public\CIP-B101_Lab1C_Ahmed_Garba>cd \Users\Public\CIP-B101_Lab1C_Ahmed_Garba\workspace
C:\Users\Public\CIP-B101_Lab1C_Ahmed_Garba\workspace>echo hello world > my_first_file.txt
C:\Users\Public\CIP-B101_Lab1C_Ahmed_Garba\workspace>type my_first_file.txt
hello world

C:\Users\Public\CIP-B101_Lab1C_Ahmed_Garba\workspace>echo Student Name: Ahmed Garba > evidence_note.txt
C:\Users\Public\CIP-B101_Lab1C_Ahmed_Garba\workspace>echo Course: CIP-B101 Lab 1C >> evidence_note.txt
C:\Users\Public\CIP-B101_Lab1C_Ahmed_Garba\workspace>echo Purpose: Windows command line forensic basics >> evidence_note.txt
C:\Users\Public\CIP-B101_Lab1C_Ahmed_Garba\workspace>type evidence_note.txt
Student Name: Ahmed Garba
Course: CIP-B101 Lab 1C
Purpose: Windows command line forensic basics
```

Task 3: Safe File and Folder Handling (Copy, Rename, and Delete)

Controlled duplication, backup, renaming, and selective deletion were executed to simulate careful evidence handling protocols.

```
C:\Users\Public\CIP-B101_Lab1C_Ahmed_Garba\workspace>:: Copy file to workspace2
C:\Users\Public\CIP-B101_Lab1C_Ahmed_Garba\workspace>cd \Users\Public\CIP-B101_Lab1C_Ahmed_Garba\workspace

C:\Users\Public\CIP-B101_Lab1C_Ahmed_Garba\workspace>copy evidence_note.txt ..\workspace2\evidence_note_copy.txt
1 file(s) copied.

C:\Users\Public\CIP-B101_Lab1C_Ahmed_Garba\workspace>
C:\Users\Public\CIP-B101_Lab1C_Ahmed_Garba\workspace>:: Recursive folder backup using xcopy
C:\Users\Public\CIP-B101_Lab1C_Ahmed_Garba\workspace>cd \Users\Public\CIP-B101_Lab1C_Ahmed_Garba

C:\Users\Public\CIP-B101_Lab1C_Ahmed_Garba>xcopy workspace workspace3\workspace_backup /E /I /H
workspace\evidence_note.txt
workspace\my_first_file.txt
2 File(s) copied

C:\Users\Public\CIP-B101_Lab1C_Ahmed_Garba>
C:\Users\Public\CIP-B101_Lab1C_Ahmed_Garba>:: Rename copied file
C:\Users\Public\CIP-B101_Lab1C_Ahmed_Garba>cd \Users\Public\CIP-B101_Lab1C_Ahmed_Garba\workspace2

C:\Users\Public\CIP-B101_Lab1C_Ahmed_Garba\workspace2>ren evidence_note_copy.txt renamed_evidence_note.txt

C:\Users\Public\CIP-B101_Lab1C_Ahmed_Garba\workspace2>
C:\Users\Public\CIP-B101_Lab1C_Ahmed_Garba\workspace2>:: Safe cleanup of temporary test files
C:\Users\Public\CIP-B101_Lab1C_Ahmed_Garba\workspace2>echo temporary test file > delete_me.txt

C:\Users\Public\CIP-B101_Lab1C_Ahmed_Garba\workspace2>del delete_me.txt

C:\Users\Public\CIP-B101_Lab1C_Ahmed_Garba\workspace2>
C:\Users\Public\CIP-B101_Lab1C_Ahmed_Garba\workspace2>:: Remove empty test folder
```

Task 4: Understanding Windows Search Pathways (PATH Variable)

System executable lookup behaviors were examined to understand how the operating system locates binaries and why PATH manipulation is relevant during forensic triage.

```
C:\Users\Public\CIP-B101_Lab1C_Ahmed_Garba>der
'der' is not recognized as an internal or external command,
operable program or batch file.

C:\Users\Public\CIP-B101_Lab1C_Ahmed_Garba>cd \Users\Public\CIP-B101_Lab1C_Ahmed_Garba

C:\Users\Public\CIP-B101_Lab1C_Ahmed_Garba>mkdir empty_test_folder
A subdirectory or file empty_test_folder already exists.

C:\Users\Public\CIP-B101_Lab1C_Ahmed_Garba>rmdir empty_test_folder

C:\Users\Public\CIP-B101_Lab1C_Ahmed_Garba>echo %PATH%
C:\Program Files (x86)\Common Files\Oracle\Java\java8path;C:\Program Files (x86)\Common Files\
WINDOWS\system32;C:\WINDOWS;C:\WINDOWS\System32\Wbem;C:\WINDOWS\System32\WindowsPowerShell\v1.0\
SH\;C:\Program Files\nodejs\;C:\Users\Ahmed\AppData\Local\Microsoft\WindowsApps;C:\Users\Ahmed\
ers\Ahmed\AppData\Local\Python\bin;C:\Users\Ahmed\AppData\Local\Programs\Microsoft VS Code\bin;

C:\Users\Public\CIP-B101_Lab1C_Ahmed_Garba>where notepad
C:\Windows\System32\notepad.exe
C:\Windows\notepad.exe
C:\Users\Ahmed\AppData\Local\Microsoft\WindowsApps\notepad.exe

C:\Users\Public\CIP-B101_Lab1C_Ahmed_Garba>notepad

C:\Users\Public\CIP-B101_Lab1C_Ahmed_Garba>vol C:
Volume in drive C has no label.
Volume Serial Number is F4A3-65CC

C:\Users\Public\CIP-B101_Lab1C_Ahmed_Garba>cls
```

```
CLIENT_RANDOM
5e85016c2010478311178455b55ee4c6cfd4fee8ef9
41b1a3fa07c3b3f86365c
d6b14d253ebe0fb069185666afc8ca76216b1b288bc
85b3c6e01a736e757f0ca4224693653d0e50ef5495f
43e1910449
CLIENT_RANDOM
5e850177170b9e2362c1c72d0dbc4a9133441079bf5
ce0cbe2909016d120967a
63c841f68786288e84a10dcfaeca8de9081e4b74a2
e1362eef50a72e92f8921fc3f4d0a38d16fcaedd883
a5af91767d
CLIENT_RANDOM
5e850178c6e06ad9d7eb8ac04fea8d4d4e5f58fe300
8dbc795cd46688a85c921
62d5fdb980a621fd5b9345f02e0758f45f8a34a6a0c
db55354306aa0f4037867124bd70d3d9c0bc743472c
c9c547ee8e
CLIENT_RANDOM
5e8501dceb9af936b5e346dc4ac6665f8dc610b37d1
a24386fe1a4d67298cbe3
f1b55a36c27c690523b401f4d8e50215ff017d0c091
3cd655bdf59d8fa56956616d9c699f1078a2b528c0
d458c3c251
CLIENT_RANDOM
5e85026cfd7611ba445bcdde52a958fe86a531cc3e8
...
Ln 1, Col 1 | 1,408 chara | Plain t | 100% | Windor | UTF-8
```

Task 5: System Information Collection

Host configuration artifacts commonly required during initial triage documentation were harvested and stored in the outputs directory.

```

C:\Users\Public\CIP-B101_Lab1C_Ahmed_Garba\outputs>systeminfo > systeminfo_full.txt

C:\Users\Public\CIP-B101_Lab1C_Ahmed_Garba\outputs>systeminfo | find "OS Name"
OS Name:                Microsoft Windows 11 Home

C:\Users\Public\CIP-B101_Lab1C_Ahmed_Garba\outputs>systeminfo | find "OS Version"
OS Version:             10.0.26200 N/A Build 26200
BIOS Version:            American Megatrends International, LLC. E1504FA.314, 4/11/2025

C:\Users\Public\CIP-B101_Lab1C_Ahmed_Garba\outputs>systeminfo | find "System Type"
System Type:             x64-based PC

C:\Users\Public\CIP-B101_Lab1C_Ahmed_Garba\outputs>systeminfo | find "Total Physical Memory"
Total Physical Memory:   15,648 MB

C:\Users\Public\CIP-B101_Lab1C_Ahmed_Garba\outputs>wmic cpu get name > cpu_info.txt
'wmic' is not recognized as an internal or external command,
operable program or batch file.

C:\Users\Public\CIP-B101_Lab1C_Ahmed_Garba\outputs>wmic diskdrive get name,model,size > disk_info.txt

```

Task 6: Network Information & Connectivity Collection

IP configurations, gateway mappings, MAC address enumerations, and loopback/external connectivity checks were performed.

```

IPv4 Address. . . . . : 192.168.100.163

C:\Users\Public\CIP-B101_Lab1C_Ahmed_Garba\outputs>ipconfig | findstr Gateway
Default Gateway . . . . . :
Default Gateway . . . . . :
Default Gateway . . . . . : 192.168.100.1

C:\Users\Public\CIP-B101_Lab1C_Ahmed_Garba\outputs>wmic nic get macaddress,description > mac_addresses.txt
'wmic' is not recognized as an internal or external command,
operable program or batch file.

C:\Users\Public\CIP-B101_Lab1C_Ahmed_Garba\outputs>ping 127.0.0.1

Pinging 127.0.0.1 with 32 bytes of data:
Reply from 127.0.0.1: bytes=32 time<1ms TTL=128
Reply from 127.0.0.1: bytes=32 time<1ms TTL=128
Reply from 127.0.0.1: bytes=32 time<1ms TTL=128
Reply from 127.0.0.1: bytes=32 time<1ms TTL=128

Ping statistics for 127.0.0.1:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 0ms, Average = 0ms

C:\Users\Public\CIP-B101_Lab1C_Ahmed_Garba\outputs>ping 8.8.8.8

Pinging 8.8.8.8 with 32 bytes of data:
Reply from 8.8.8.8: bytes=32 time=26ms TTL=115
Reply from 8.8.8.8: bytes=32 time=27ms TTL=115

```

Task 7: Searching Text Files for Evidence (findstr)

A mini case log was generated to simulate searching parsed artifact files for specific indicators of compromise or keywords.

```

C:\Users\Public\CIP-B101_Lab1C_Ahmed_Garba\outputs>ping google.comcd \Users\Public\CIP-B101_Lab1C_Ahmed_Garba\evidence
Ping request could not find host google.comcd. Please check the name and try again.

C:\Users\Public\CIP-B101_Lab1C_Ahmed_Garba\outputs>echo User: Ahmed Garba > mini_case_log.txt

C:\Users\Public\CIP-B101_Lab1C_Ahmed_Garba\outputs>echo Email: trainee@example.com >> mini_case_log.txt

C:\Users\Public\CIP-B101_Lab1C_Ahmed_Garba\outputs>echo IP: 192.168.56.10 >> mini_case_log.txt

C:\Users\Public\CIP-B101_Lab1C_Ahmed_Garba\outputs>echo Keyword: suspicious_login >> mini_case_log.txt

C:\Users\Public\CIP-B101_Lab1C_Ahmed_Garba\outputs>
C:\Users\Public\CIP-B101_Lab1C_Ahmed_Garba\outputs>findstr /i "email" mini_case_log.txt
Email: trainee@example.com

C:\Users\Public\CIP-B101_Lab1C_Ahmed_Garba\outputs>findstr /i "suspicious" mini_case_log.txt
Keyword: suspicious_login

C:\Users\Public\CIP-B101_Lab1C_Ahmed_Garba\outputs>findstr /n /i "ip" mini_case_log.txt

```

Task 8: File Ownership and Permissions Check

File metadata, ownership, and discretionary access control lists (DACCs) were examined.

```
C:\Users\Public\CIP-B101_Lab1C_Ahmed_Garba\outputs>findstr /n /i "ip" mini_case_log.txtcd \Users\Public\CIP-B101_Lab1C_Ahmed_Garba\outputs
FINDSTR: Cannot open mini_case_log.txtcd

C:\Users\Public\CIP-B101_Lab1C_Ahmed_Garba\outputs>dir /q
Volume in drive C has no label.
Volume Serial Number is F4A3-65CC

Directory of C:\Users\Public\CIP-B101_Lab1C_Ahmed_Garba\outputs

08/09/2026  05:05 PM  <DIR>          ERROR404\garba  .
08/09/2026  04:58 PM  <DIR>          ERROR404\garba  ..
08/09/2026  05:03 PM                0 ERROR404\garba  cpu_info.txt
08/09/2026  05:04 PM                0 ERROR404\garba  disk_info.txt
08/09/2026  05:04 PM            3,921 ERROR404\garba  ipconfig_all.txt
08/09/2026  05:04 PM                0 ERROR404\garba  mac_addresses.txt
08/09/2026  05:05 PM                97 ERROR404\garba  mini_case_log.txt
08/09/2026  05:03 PM            4,514 ERROR404\garba  systeminfo_full.txt
               6 File(s)              8,532 bytes
               2 Dir(s)  140,563,951,616 bytes free

C:\Users\Public\CIP-B101_Lab1C_Ahmed_Garba\outputs>icacls mini_case_log.txt
```

Task 9: Process Listing and Browser Process Identification

Running processes were queried to identify active user applications and background execution states.

```
Directory of C:\Users\Public\CIP-B101_Lab1C_Ahmed_Garba\outputs

08/09/2026  05:05 PM  <DIR>          ERROR404\garba  .
08/09/2026  04:58 PM  <DIR>          ERROR404\garba  ..
08/09/2026  05:03 PM                0 ERROR404\garba  cpu_info.txt
08/09/2026  05:04 PM                0 ERROR404\garba  disk_info.txt
08/09/2026  05:04 PM            3,921 ERROR404\garba  ipconfig_all.txt
08/09/2026  05:04 PM                0 ERROR404\garba  mac_addresses.txt
08/09/2026  05:05 PM                97 ERROR404\garba  mini_case_log.txt
08/09/2026  05:03 PM            4,514 ERROR404\garba  systeminfo_full.txt
               6 File(s)              8,532 bytes
               2 Dir(s)  140,563,951,616 bytes free

C:\Users\Public\CIP-B101_Lab1C_Ahmed_Garba\outputs>icacls mini_case_log.txtcd \Users\Public\CIP-B101_Lab1C_Ahmed_Garba\o
utputs
Invalid parameter "\Users\Public\CIP-B101_Lab1C_Ahmed_Garba\outputs"

C:\Users\Public\CIP-B101_Lab1C_Ahmed_Garba\outputs>wmic process list brief > process_list.txt
'wmic' is not recognized as an internal or external command,
operable program or batch file.

C:\Users\Public\CIP-B101_Lab1C_Ahmed_Garba\outputs>wmic process where "name like '%chrome%'" get name,processid,commandl
ine
'wmic' is not recognized as an internal or external command,
operable program or batch file.

C:\Users\Public\CIP-B101_Lab1C_Ahmed_Garba\outputs>wmic process where "name like '%msedge%'" get name,processid,commandl
ine
```

Task 10: Batch File Creation and Execution

10.1 hello_world.bat

Created inside the scripts folder to verify execution capability:

```
C:\Users\Public\CIP-B101_Lab1C_Ahmed_Garba\outputs>wmic process where "name like '%msedge%'" get name,processid,commandline@ECHO OFF
'wmic' is not recognized as an internal or external command,
operable program or batch file.

C:\Users\Public\CIP-B101_Lab1C_Ahmed_Garba\outputs>ECHO Hello World!
Hello World!

C:\Users\Public\CIP-B101_Lab1C_Ahmed_Garba\outputs>PAUSE
```

10.2 sys_info.bat

Developed as a reusable triage automation script:

```
C:\Users\Public\CIP-B101_Lab1C_Ahmed_Garba\outputs>wmic diskdrive get name,model,size >> ..\outputs\sys_info_report.txt
'wmic' is not recognized as an internal or external command,
operable program or batch file.

C:\Users\Public\CIP-B101_Lab1C_Ahmed_Garba\outputs>ECHO. >> ..\outputs\sys_info_report.txt

C:\Users\Public\CIP-B101_Lab1C_Ahmed_Garba\outputs>ECHO ===== >> ..\outputs\sys_info_report.txt

C:\Users\Public\CIP-B101_Lab1C_Ahmed_Garba\outputs>ECHO NETWORK INFO >> ..\outputs\sys_info_report.txt

C:\Users\Public\CIP-B101_Lab1C_Ahmed_Garba\outputs>ECHO ===== >> ..\outputs\sys_info_report.txt

C:\Users\Public\CIP-B101_Lab1C_Ahmed_Garba\outputs>ipconfig | findstr IPv4 >> ..\outputs\sys_info_report.txt

C:\Users\Public\CIP-B101_Lab1C_Ahmed_Garba\outputs>wmic nic get macaddress,description >> ..\outputs\sys_info_report.txt

'wmic' is not recognized as an internal or external command,
operable program or batch file.

C:\Users\Public\CIP-B101_Lab1C_Ahmed_Garba\outputs>START https://support.microsoft.com/en-us/windows/windows-10-system-requirements-6d4e9a79-66bf-7950-467c-795cf0386715

C:\Users\Public\CIP-B101_Lab1C_Ahmed_Garba\outputs>ECHO Collection complete. See outputs\sys_info_report.txt
Collection complete. See outputs\sys_info_report.txt

C:\Users\Public\CIP-B101_Lab1C_Ahmed_Garba\outputs>PAUSE
```

Task 11: Making sys_info.bat Globally Callable via PATH

To ensure scripts can be executed from arbitrary directories without specifying absolute file paths, the environment PATH variable was temporarily updated:

```
C:\Users\Public\CIP-B101_Lab1C_Ahmed_Garba\outputs>ECHO ===== >> ..\outputs\sys_info_report.txt

C:\Users\Public\CIP-B101_Lab1C_Ahmed_Garba\outputs>ECHO NETWORK INFO >> ..\outputs\sys_info_report.txt

C:\Users\Public\CIP-B101_Lab1C_Ahmed_Garba\outputs>ECHO ===== >> ..\outputs\sys_info_report.txt

C:\Users\Public\CIP-B101_Lab1C_Ahmed_Garba\outputs>ipconfig | findstr IPv4 >> ..\outputs\sys_info_report.txt

C:\Users\Public\CIP-B101_Lab1C_Ahmed_Garba\outputs>wmic nic get macaddress,description >> ..\outputs\sys_info_report.txt

'wmic' is not recognized as an internal or external command,
operable program or batch file.

C:\Users\Public\CIP-B101_Lab1C_Ahmed_Garba\outputs>START https://support.microsoft.com/en-us/windows/windows-10-system-requirements-6d4e9a79-66bf-7950-467c-795cf0386715

C:\Users\Public\CIP-B101_Lab1C_Ahmed_Garba\outputs>ECHO Collection complete. See outputs\sys_info_report.txt
Collection complete. See outputs\sys_info_report.txt

C:\Users\Public\CIP-B101_Lab1C_Ahmed_Garba\outputs>cd \Users\Public\CIP-B101_Lab1C_Ahmed_Garba

C:\Users\Public\CIP-B101_Lab1C_Ahmed_Garba>copy scripts\sys_info.bat workspace4\sys_info.bat
The system cannot find the file specified.

C:\Users\Public\CIP-B101_Lab1C_Ahmed_Garba>set PATH=%PATH%;C:\Users\Public\CIP-B101_Lab1C_Ahmed_Garba\workspace4

C:\Users\Public\CIP-B101_Lab1C_Ahmed_Garba>cd \

C:\>sys_info.bat|
```

What is the difference between an absolute path and a relative path? Give one example of each from your lab.

Absolute Path: Specifies the complete location of a file or directory starting from the root drive (e.g., C:\Users\Public\CIP-B101_Lab1C_Ahmed_Garba).

Relative Path: Specifies a location relative to the current working directory (e.g., ..\workspace2\evidence_note_copy.txt).

Why should analysts create a separate lab/evidence folder before running commands?

Creating an isolated workspace prevents accidental modification, overwriting, or deletion of critical operating system files or pre-existing case artifacts, ensuring strict chain-of-custody and organization.

Which command showed your IP address and which command showed your MAC address?

IP address information was retrieved using ipconfig (or ipconfig | findstr IPv4).

MAC address information was retrieved using wmic nic get macaddress,description.

Why is a batch file useful in forensic triage?

Batch scripts automate repetitive, multi-step triage procedures, minimizing human error, standardizing data collection across multiple workstations, and accelerating evidence acquisition during time-sensitive investigations.

Why must analysts document failed commands or failed pings instead of hiding them?

Complete transparency is vital for forensic integrity. Failed pings or errors provide crucial environmental context (e.g., network isolation, disconnected adapters, or missing privileges) that investigators and reviewers must account for.

What risk can occur if a suspicious executable is placed in a directory listed in PATH?

If a malicious binary shares the name of a standard utility (or relies on PATH hijacking/DLL preloading vectors), the operating system may inadvertently execute the unauthorized program when a user or script invokes the command without a fully qualified path.

CONCLUSION

This practical laboratory successfully reinforced core Windows command-line operations essential for digital forensics. By establishing structured workspaces, executing system and network triage queries, automating artifact collection with batch scripts, and managing execution paths, foundational technical proficiency and documentation discipline necessary for advanced forensic investigations were achieved.